Day 301 – Deep Learning Theory

What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple layers to learn patterns from data. These neural networks are inspired by the structure and functioning of the human brain.

Traditional Machine Learning often requires manual feature extraction, while Deep Learning can automatically learn features directly from raw data such as images, text, audio, and videos.

Deep Learning is widely used in:

- Image Recognition
- Speech Recognition
- Natural Language Processing
- Recommendation Systems
- Autonomous Vehicles
- Medical Diagnosis
- Fraud Detection

Artificial Intelligence vs Machine Learning vs Deep Learning

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Artificial Intelligence (AI)
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├── Machine Learning (ML)
|   |
|   ├── Supervised Learning
|   ├── Unsupervised Learning
|   └── Reinforcement Learning
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└── Deep Learning (DL)
    |
    ├── Neural Networks
    ├── CNN = Convolutional Neural Network
    ├── RNN = Recurrent Neural Network
    └── Transformers
```

Artificial Intelligence (AI)

AI refers to systems capable of performing tasks that normally require human intelligence.

Examples:

- Chatbots
- Virtual Assistants
- Expert Systems

Machine Learning (ML)

Machine Learning is a subset of AI where systems learn patterns from data and make predictions.

Examples:

- House Price Prediction
- Spam Detection
- Customer Churn Prediction

Deep Learning (DL)

Deep Learning is a subset of Machine Learning that uses multiple neural network layers to learn complex patterns.

Examples:

- Face Recognition
- Self-Driving Cars
- Language Translation
- ChatGPT-like Systems

Why Deep Learning?

Traditional Machine Learning struggles with:

- Large datasets
- Images
- Videos
- Audio
- Natural language

Deep Learning can:

- Learn complex relationships
- Process unstructured data
- Improve performance with more data
- Reduce manual feature engineering

What is a Neural Network?

A Neural Network is a computational model inspired by biological neurons.

It consists of interconnected nodes called neurons.

Each neuron receives input, processes it, and produces an output.

Basic Neural Network Structure

Input Layer

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Hidden Layer

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Output Layer

Layers in Neural Networks

Input Layer

The input layer receives raw data.

Example:

Predicting House Price:

Area

Bedrooms

Bathrooms

Location Score

These values become inputs to the network.

Hidden Layer

The hidden layer performs computations and learns patterns.

A neural network may contain:

- One hidden layer
- Multiple hidden layers

Deep Learning typically uses many hidden layers.

Output Layer

The output layer generates the final prediction.

Examples:

Binary Classification:

Spam

Not Spam

House Price Prediction:

₹ 45,00,000

What is a Neuron?

A neuron is the basic building block of a neural network.

A neuron:

1. Receives inputs
2. Applies weights
3. Adds bias
4. Applies activation function
5. Produces output

Neuron Structure

```
Input1 ──┐
         │
Input2 ──┼──▶ Weighted Sum
         │
Input3 ──┘
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+ Bias

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Activation Function

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Output

Inputs

Inputs are features given to the model.

Example:

Student Performance Prediction

Study Hours

Attendance

Assignments Completed

Inputs:

x1

x2

x3

What are Weights?

Weights determine the importance of each input.

Notation:

w1

w2

w3

Example:

Study Hours = Important

Attendance = Moderate

Assignments = Less Important

Weights help the model understand this importance.

Example

Inputs:

x1 = 5

x2 = 3

Weights:

w1 = 0.8

w2 = 0.2

Weighted Sum:

(5 × 0.8) + (3 × 0.2)

= 4 + 0.6

= 4.6

What is Bias?

Bias is an additional value added to the weighted sum.

Purpose:

- Shift output
- Improve learning flexibility
- Help fit data better

Formula:

Output = Weighted Sum + Bias

Example:

4.6 + 1.0

= 5.6

Why is Bias Important?

Without bias:

Output always passes through origin

With bias:

Output can shift freely

This improves model performance.

Mathematical Representation of a Neuron

Formula:

$z = (w1x1 + w2x2 + w3x3) + b$

Where:

z = weighted sum

w = weights

x = inputs

b = bias

After activation:

a = Activation(z)

Activation Function

Activation functions introduce non-linearity.

Without activation functions:

Neural Network = Linear Regression

Deep Learning becomes powerful because of activation functions.

Examples:

- ReLU
- Sigmoid
- Tanh
- Softmax

(Detailed study on Day 305)

Forward Propagation

Forward Propagation is the process of moving data from:

Input Layer

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Hidden Layer

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Output Layer

Steps:

1. Input received
2. Multiply by weights
3. Add bias
4. Apply activation
5. Produce output

Example Forward Pass

Input = 10

Weight = 0.5

Bias = 1

Calculation:

$(10 \times 0.5) + 1$

$= 6$

Output:

6

What is Loss Function?

The Loss Function measures how wrong the prediction is.

Formula:

Loss = Actual Value - Predicted Value

Smaller loss:

Better Model

Larger loss:

Poor Model

Example

Actual House Price:

₹50,00,000

Predicted:

₹45,00,000

Error:

₹5,00,000

Loss Function measures this error.

Common Loss Functions

Mean Squared Error (MSE)

Used in Regression.

Formula:

MSE = $\Sigma(\text{actual} - \text{predicted})^2 / n$

Binary Cross Entropy

Used in Binary Classification.

Example:

Spam

Not Spam

Categorical Cross Entropy

Used in Multi-Class Classification.

Example:

Digit Recognition

0-9

What is an Epoch?

An Epoch means one complete pass through the entire training dataset.

Example:

Dataset:

1000 records

Epoch:

Model sees all 1000 records once

Example

Epoch 1:

Train on all data

Epoch 2:

Train on all data again

Epoch 3:

Train on all data again

Why Multiple Epochs?

A single pass is usually insufficient.

The model gradually learns.

Example:

Epoch 1

Accuracy = 55%

Epoch 10

Accuracy = 82%

Epoch 20

Accuracy = 92%

What Happens During Training?

Input Data

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Forward Propagation

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Prediction

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Loss Function

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Calculate Error

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Update Weights

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Next Epoch

Deep Learning Training Workflow

Dataset

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Input Layer

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Hidden Layers

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Output Layer

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Prediction

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Loss Function

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Backpropagation

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Weight Update

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Next Epoch

Advantages of Deep Learning

- Automatic feature extraction
- Handles large datasets
- High accuracy
- Learns complex patterns
- Excellent for images and text
- Scalable

Disadvantages of Deep Learning

- Requires large datasets
- Requires powerful hardware
- Long training times
- Difficult to interpret
- High computational cost

Real-World Applications

Computer Vision

- Face Recognition
- Object Detection
- Medical Imaging

Natural Language Processing

- Chatbots
- Language Translation
- Sentiment Analysis

Speech Recognition

- Voice Assistants
- Voice Commands
- Speech-to-Text

Recommendation Systems

- Netflix
- Amazon
- YouTube

Autonomous Vehicles

- Lane Detection
- Traffic Sign Recognition
- Obstacle Detection

Key Terms Summary

Term	Meaning
Neural Network	Collection of interconnected neurons
Neuron	Basic processing unit
Weight	Importance of an input
Bias	Extra value added to weighted sum
Activation Function	Introduces non-linearity
Forward Propagation	Data moving through network
Loss Function	Measures prediction error
Epoch	One complete pass through training data
Deep Learning	Multi-layer neural networks for complex learning

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Day 301 Summary

In this lesson, you learned:

- What Deep Learning is
- Difference between AI, ML, and DL
- Neural Networks
- Neurons
- Inputs
- Weights
- Biases
- Activation Functions
- Forward Propagation
- Loss Functions
- Epochs
- Deep Learning Training Workflow

These concepts form the foundation for all upcoming Deep Learning topics, including TensorFlow, Keras, CNNs, Transfer Learning, NLP, and Deep Learning web applications.